Fluorescence Microscopy: Cellular Structure

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Onion epidermal cells and human cheek epithelial cells were stained with Hoechst dye and imaged in brightfield and UV-excited fluorescence at 20× (NA 0.45). In brightfield, the onion epidermis is a continuous sheet of elongated cells; eight complete cells are 290–520 µm long and 47–76 µm wide. Cheek cells appear as a few faint, flat polygons among bubbles and debris. Fluorescence identified one nucleus per cell: 29 µm in median diameter in onion (9 nuclei) and 8 µm in cheek cells (10 nuclei), at 1.6–5 times the local background in the green channel. The fluorescence background was bright blue rather than dark, so the contrast came mainly from the green channel; its source was not identified. A longer exposure raised the background more than the nuclear signal.

1. INTRODUCTION

Most cellular structures are nearly transparent and differ little in refractive index, so in brightfield illumination they show little contrast. Fluorescence microscopy instead images a dye that binds selectively to one kind of molecule: the dye is excited at one wavelength, and filters pass only its longer-wavelength emission to the detector [1,2].

Hoechst dyes bind to the minor groove of double-stranded DNA, preferentially at A–T-rich sequences, and so mark cell nuclei. Bound dye absorbs near 350 nm and emits near 461 nm [2]. The dyes cross intact membranes, so they stain freshly prepared cells within minutes.

This report compares brightfield and Hoechst fluorescence images of two cell types: onion epidermal cells (plant) and human cheek epithelial cells (animal). The manual asks for a comparison of their structure, size and morphology, and of what fluorescence adds to brightfield imaging [1].

2. THEORY

2.1 Fluorescence and the Stokes Shift

A fluorophore absorbs a photon and is raised from the ground state S_0 to a vibrationally excited level of the first excited singlet state S_1. It relaxes non-radiatively to the lowest level of S_1 and then returns to S_0 by emitting a photon [2]. The energy lost in relaxation makes the emitted photon longer in wavelength than the absorbed one. This difference, the Stokes shift, allows filters to separate the weak emission from the much stronger excitation light. Hoechst becomes much more fluorescent on binding DNA, because the rigid groove suppresses non-radiative decay [2]. Unbound dye therefore contributes little, but not nothing, to the background.

2.2 Epi-Illumination and Filter Cubes

In epi-illumination the objective both delivers the excitation light and collects the emission [1]. A filter cube holds an excitation filter, a dichroic mirror that reflects the excitation light towards the specimen and transmits the emission, and a barrier filter that blocks residual excitation. The microscope has three cubes, for UV (WU), blue (WB) and green (WG) excitation [1]. Hoechst requires the WU cube, which excites at about 330–385 nm and passes all emission above about 420 nm. Because this barrier filter is a long-pass, any emission above 420 nm reaches the camera, not only that of the bound dye.

2.3 Resolution, Sampling and Depth of Field

By the Rayleigh criterion, the smallest resolvable separation is [3]

$$d_{\min} = \frac{0.61\lambda}{\mathrm{NA}}, \tag{1}$$

where λ is the emission wavelength and NA the numerical aperture of the objective. For NA = 0.45 and $\lambda \approx 461$ nm, $d_{\min} \approx 0.63$ µm. This is about one camera pixel (0.599 µm). Sampling requires at least two pixels per resolved distance, so the camera, not diffraction, limits the image resolution to about 1.2 µm. The depth of field, approximately $\lambda/\mathrm{NA}^2 \approx 2$ µm for the optics alone, is a few micrometres. Structures more than a few micrometres from the focal plane therefore appear blurred.

3. METHODS

3.1 Sample Preparation

Samples were prepared following the manual [1]. For the onion cells, the thin transparent inner membrane of a scale leaf was peeled off with forceps, stained with Hoechst dye for 15 min, and placed on a slide under a coverslip. For the cheek cells, the inner lining of the cheek was gently scraped with the flat end of a toothpick. The cells were transferred to a drop of PBS on a slide, stained with Hoechst for 15 min and covered with a coverslip. The dye concentration and variant (33258 or 33342) were not recorded.

3.2 Imaging

Images were taken on an Olympus microscope with a 20× air objective (NA 0.45) and an Olympus DP74 colour camera controlled by the DP Controller software. Each brightfield image was taken with the transmitted-light source. For fluorescence, the transmitted light was switched off, the WU cube selected and the shutter opened. Exposure was set manually for each image, and the gain was 5 or 6 (Table I).

The pixel size is 0.599 µm, from the calibrated magnification of 19.56 and a 0.5× camera adapter. This value is used for all scale bars and measurements below, and gives a field of view of 959 × 719 µm for the 1600 × 1200 pixel images.

TABLE I. Exposure time and gain of each image, in the order taken, all at 20× (NA 0.45). Each field (cheek C1, C2; onion O1, O2) was imaged in brightfield and in fluorescence.

Field	Cells	Mode	Exp. (ms)	Gain
C1	Cheek	Fluor.	105.0	5
C1	Cheek	Bright.	1.3	5
C2	Cheek	Fluor.	32.4	6
C2	Cheek	Bright.	0.61	6
O1	Onion	Bright.	0.76	6
O1	Onion	Fluor.	23.2	6
O2	Onion	Bright.	0.76	6
O2	Onion	Fluor.	42.2	6

3.3 Image Analysis

All measurements were made with a Python script on the raw images. The onion cells were measured in the brightfield images of fields O1 and O2. Walls were taken as the pixels darker than 80% of the local mean brightness and were widened by 4 pixels to close small gaps. Each region enclosed by walls and not touching the image edge was taken as one complete cell. The length of a cell is its extent along its long axis, and its mean width is its area divided by its length. Because the walls were widened, the widths are underestimated by roughly 8 µm. Cells that extend beyond the field are excluded, which biases the lengths low.

Nuclei were located in the green channel, in which they stand out; the blue channel is close to its maximum value across the whole field. A nucleus was taken as a region more than 25 robust standard deviations above the smoothed background. Its diameter is that of a circle with the area above half the nuclear peak, and its contrast is the peak green value divided by the local background.

4. RESULTS

4.1 Onion Epidermal Cells

In brightfield (Fig. 1(a)), the onion epidermis is a continuous sheet of elongated cells, joined end to end in staggered rows and bounded by dark walls. The eight complete cells found in fields O1 and O2 are 290–520 µm long (median 410 µm) and 47–76 µm wide (median 66 µm), so a typical cell is about six times longer than it is wide. Nuclei are hard to pick out in brightfield.

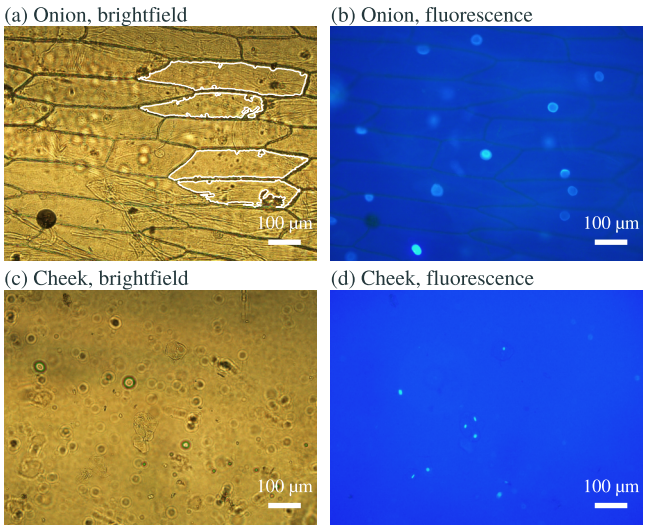


FIG. 1. Onion epidermal cells (a, b; field O1) and cheek epithelial cells (c, d; field C2), in brightfield (left) and Hoechst fluorescence (right) at 20×. Each row shows one field. White outlines in (a) mark the four complete cells measured in this image. Scale bars are 100 µm.

In fluorescence (Fig. 1(b)), nuclei appear as bright cyan disks, at most one per cell. The nine nuclei found in field O1 have diameters of 15–36 µm (median 29 µm) and are 2.2–5.2 times brighter than the surrounding background in the green channel. Several appear as rings or as blurred disks, consistent with nuclei lying outside the few-micrometre depth of field. The background is not dark: it is a uniform blue, against which the cell walls show as darker lines. Field O2 is similar but more blurred.

4.2 Cheek Epithelial Cells

In brightfield (Fig. 1(c)), only a few cheek cells can be seen, as thin, flat, irregular polygons with faint outlines. Measured by eye against the scale bar, the two clearest are about 50 and 60–90 µm across. Most of the round objects with bright rings are not cells: they are much smaller than squamous cells, have sharp diffraction rings and are not stained, so they are most likely air bubbles or debris.

In fluorescence (Fig. 1(d)), the ten nuclei found in field C2 are small, compact spots 5–14 µm across (median 8 µm) and are 1.6–3.7 times brighter than the background. Several lie close together near the centre, where a cluster of cells is faintly visible in brightfield. The cytoplasm and outlines are barely visible in fluorescence.

4.3 Exposure and Background

In all fluorescence images, the background is bright: the median blue value is 240–246 out of 255 in fields C1, C2 and O2, and 186 in field O1. Fewer than 0.02% of pixels reach 255, so the camera is not saturated. Instead the blue channel lies within a few levels of its maximum everywhere, so it carries almost no image contrast. The nuclei are separated from the background mainly in the green channel.

The two cheek fields were imaged in fluorescence at 105 ms with gain 5 (C1) and at 32.4 ms with gain 6 (C2). In field C1,

the green background is about twice as high (135 against 70), and the nuclear contrast is lower (median 1.6 against 3.3). The longer exposure raised the background more than the nuclear signal. Because the field and gain also differ, this comparison is qualitative.

5. DISCUSSION

5.1 Plant and Animal Cells

The two cell types differ as expected. Onion epidermal cells form a continuous sheet of cells bounded by walls, each roughly 400 × 65 µm. Cheek cells, scraped from their tissue, lie flat and isolated, are bounded only by a membrane, and are seen in brightfield as faint polygons 50–90 µm across [4]. Both contain a single nucleus per cell. The onion nuclei are about 3.5 times larger in diameter than the cheek nuclei (29 against 8 µm). The cell wall itself is not demonstrated chemically here; the inference rests on the shape and rigidity of the onion cells and on the textbook composition of plant cell walls [4].

5.2 What Fluorescence Added

Fluorescence identified the nuclei, which brightfield did not. In the cheek sample this distinction mattered most: in brightfield, the cells could hardly be separated from bubbles and debris, whereas each Hoechst-stained nucleus marks a cell. The image of the bound dye is specific to DNA: no stained structure appeared outside the nuclei. The contrast, however, was limited (nuclei 1.6–5 times the background) because the background was not dark.

5.3 Limitations of These Images

The bright blue background has several possible sources: unbound dye, autofluorescence of the specimen or the slide, and excitation light leaking through the long-pass filter. An unstained control slide imaged at the same settings would distinguish them. The camera's colour processing also affects the result, because the analysis uses the green channel of a processed colour image, not a calibrated intensity. Each comparison rests on one or two fields per specimen, the exposure and gain differed between images, and each specimen was imaged at a single focal plane, so out-of-focus onion nuclei appear larger and fainter. The cell sizes come from eight complete onion cells; no cheek cell outline was clear enough to measure automatically.

6. CONCLUSION

Brightfield images showed the onion epidermis as a sheet of elongated cells, 290–520 µm long and 47–76 µm wide, and showed few cheek cells clearly. Hoechst fluorescence identified one nucleus per cell in both: about 29 µm in diameter in onion and 8 µm in cheek cells, at 1.6–5 times the background brightness. The fluorescence background was bright rather than dark, so the nuclei stood out mainly in the green channel. Four changes would improve the measurement. Imaging an unstained control at the same settings would identify the source of the background. Fixed exposure and gain for every fluorescence image would make the images comparable. A focus series would give in-focus nuclear sizes. Mounting the cheek cells without trapped air and imaging several fields per slide would give enough cheek cells to measure their size.

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